∃ is the Image

by notes on Freyd & Scedrov, Categories · Allegories

The existential quantifier as direct image — §1.451 (inverse image), §1.6 (pre-logos)

Fix a map $f : A \to B$. It carries subsets **forward** (direct image $\exists_f$) and **backward** (inverse image $f^{\#}$); the forward one **is** the existential quantifier. We watch this in SET element by element, see why projection gives $\exists y$, and read off the adjunction $\exists_f \vDash f^{\#}$.

Remark 0.1 (Setup).
$f : A \to B$ a map. A **subset** $S \subseteq A$ is a unary predicate on A. Composition is written in **diagram order**: $S \hookrightarrow A \xrightarrow{f} B$ means "first include S, then apply f". In SET the operative element rule is: apply f to each element.

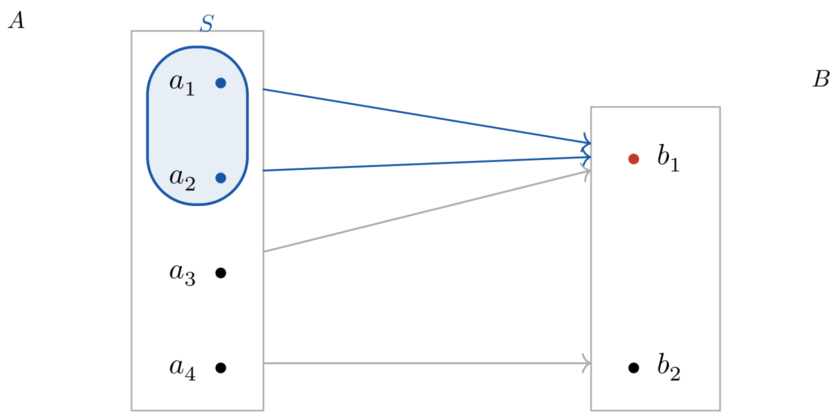
§1. Direct image, element by element

Definition 1.1 (Direct image $\exists_f$).
For $S \subseteq A$,

$$\exists_f(S) \quad = \quad f(S) \quad = \quad \{\, b \in B \mid \exists a \in S.\, f(a) = b \,\}.$$

The "$\exists a$" is **literally in the definition**: one element of S landing on b is enough.

b lies in $\exists_f(S)$ iff its **fibre** $f^{-1}(b)$ **meets** S. Below, $S = \{a_1, a_2\}$ is the blue circle.



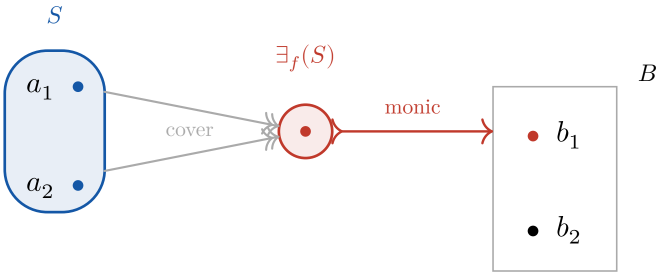
f (apply to each element)

⋆ Punchline

Meets, not contains. The fibre over b_1 is $\{a_1, a_2, a_3\}$ — it only **overlaps** S (the witness $a_3 \notin S$), yet $b_1 \in \exists_f(S)$: one witness in S is enough. The fibre over b_2 is $\{a_4\}$ and misses S entirely, so $b_2 \notin \exists_f(S)$. Requiring the **whole** fibre $\subseteq S$ would instead give the universal image $\forall_f(S)$ — the ∀ quantifier of the last section, a strictly different, dual construction.

§2. The categorical picture: image factorization

In SET the composite $S \hookrightarrow A \xrightarrow{f} B$ splits in two — collapse equal values, then include. With $S = \{a_1, a_2\}$, both sent to b_1:



The **cover** fuses a_1, a_2 onto one point; the **monic** includes that point into B as the subobject $\exists_f(S)$. A general category has no elements, but this same factorization of $S \hookrightarrow A \xrightarrow{f} B$ — cover then monic — **is** the definition:

$$\begin{tikzcd}
S \arrow[r, hook, "m_S"] \arrow[d, two heads, "\text{cover}"'] \arrow[dr, "m_S\, f"'] & A \arrow[d, "f"] \\
\exists_f(S) \arrow[r, hook, "m"] & B
\end{tikzcd}$$

Definition 2.1 (In the repo (`Fredy/S1_60.lean`)).
The diagram's monic is m_S = `S.arr` (the inclusion $S \hookrightarrow A$), so

$$\exists_f(S) \quad := \quad \mathrm{image}(m_S\, f),$$

the subobject $\exists_f(S) \hookrightarrow B$ drawn above. In Lean: `existsAlong f S := image (S.arr ≫ f)` (generic signature `existsAlong g U`). The adjunction (next section) is `existsAlong_le_iff`. Needs only a **regular** category, not the full pre-logos.

§3. Why projection gives $\exists y$

Take $f = \pi_X : X \times Y \to X$, the projection that **forgets** Y. A subset $P \subseteq X \times Y$ is a **two-place** predicate $p(x, y)$. Push it forward:

$$\exists_{\pi_X}(P) = \{\, x \in X \mid \exists (x', y) \in P.\, x' = x \,\} = \{\, x \mid \exists y.\, p(x, y) \,\}.$$

So $\exists_{\pi_X}$ along the projection that drops Y **is** the quantifier $\exists y$ — the source of the name.

$$\begin{tikzcd}
X \times Y \arrow[r, "\pi_X\ (\text{drop } y)"] \arrow[d] & X \arrow[d] \\
p(x, y) \arrow[r, mapsto, "\exists_{\pi_X}"] & \exists y.\, p(x, y)
\end{tikzcd}$$

§4. The adjunction $\exists_f \vDash f^{\#}$

Forward image is **left adjoint** to backward image — a **Galois connection** between the subset lattices:

$$\exists_f(S) \subseteq T \iff S \subseteq f^{\#(T)} \qquad (S \subseteq A,\, T \subseteq B).$$

Both sides say "S maps into T"; in Set, `all (∈ T) (map f S)` = `all (λa. f a ∈ T) S`.

$$\begin{tikzcd} \mathcal{S}\mathrm{ub}(A) \arrow[rr, bend left, "\exists_f \text{ (image)}"] \arrow[rr, phantom, "\vDash"] & & \mathcal{S}\mathrm{ub}(B) \arrow[ll, bend left, "f^{\#} \text{ (preimage)}"] \end{tikzcd}$$

Remark 4.1 (The whole hierarchy in one line).
$f^{\#}$ is the substitution in the middle of $\exists_f \vDash f^{\#} \vDash \forall_f$.

- $\exists_f \vDash f^{\#}$ — a **regular** category. Gives $\exists$ (and that $f^{\#}$ preserves $\cap$, "for free", as a right adjoint).
- $f^{\#} \vDash \forall_f$ — a **Heyting** category (Freyd's **logos**). Gives $\forall$.

The **pre-logos** (coherent) layer in between is exactly: $f^{\#}$ **also** preserves unions $\cup$ — the one preservation that is **not** free, and the content behind the §1.616 proofs in `S1_60.lean`.

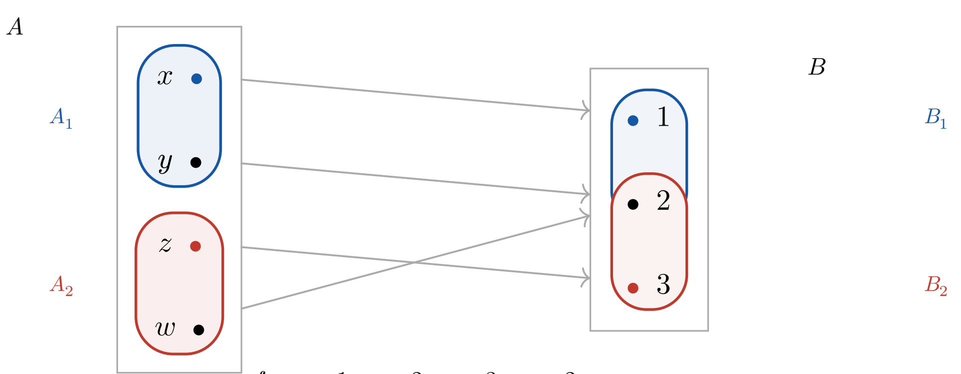
§5. The third definition: "pullbacks transfer finite covers"

Freyd gives a **third**, equivalent definition of pre-logos (§1.611) that never mentions lattices:

A **Cartesian category with images** in which **pullbacks transfer finite covers**: given a cover $\{B_i \twoheadrightarrow B\}$ and any map $A \to B$, the pullbacks $A_i = A \times_B B_i$ form a cover $\{A_i \twoheadrightarrow A\}$.

Unpacked in Set: a **cover** is a surjection, and a finite family $\{B_i \twoheadrightarrow B\}$ is a cover when it is **jointly surjective** — every $b \in B$ is hit by some B_i, i.e. as subobjects $\cup_i B_i = B$. Pulling back along f is taking the **preimage**. The claim: preimage of a jointly-surjective family is jointly surjective.

§5.1. A concrete instance



$f\colon\ x \mapsto 1,\ y \mapsto 2,\ z \mapsto 3,\ w \mapsto 2$

The cover of B is $B_1 = \{1, 2\}$, $B_2 = \{2, 3\}$ (jointly $= B$). Their preimages

$$A_1 = f^{\#(B_1)} = \{x, y, w\}, \quad A_2 = f^{\#(B_2)} = \{y, z, w\}$$

satisfy $A_1 \cup A_2 = \{x, y, z, w\} = A$ — so $\{A_1, A_2\}$ covers A. The cover **transferred down the pullback**. (Note y, w land in **both** A_i because $2 \in B_1 \cap B_2$; overlap is fine.)

§5.2. Why this is the same axiom, covering-side-up

A cover **is** the equation $B_1 \cup B_2 = B$ (the union is the top $\top_B$). "Transfers" then says

$$f^{\#(B_1 \cup B_2)} = f^{\#(B_1)} \cup f^{\#(B_2)} \qquad (= \top_A),$$

which is just $f^{\#}$ **preserving unions** — Definition 2 — in the special case where the union is the whole object. Relativizing $\top$ to an arbitrary subobject recovers the general statement. So Definitions 2 and 3 are the **same** condition, stated bottom-up (subobjects) vs. top-down (covers).

⋆ **Punchline**

In Set it is automatic; in general it is a real axiom. The one-line proof: for $a \in A$, $f(a)$ is hit by some B_i, so $a \in f^{\#(B_i)}$ — preimage of jointly-surjective is jointly-surjective. This works because Set has **extensive coproducts** (f^{-1} commutes with disjoint union), i.e. it is **positive** (§1.623). In a bare regular category the pullback squares always **exist**, but the forward inclusion $f^{\#(B_1 \cup B_2)} \leq f^{\#(B_1)} \cup f^{\#(B_2)}$ can fail — which is exactly why Freyd states "transfer finite covers" as a **hypothesis**, and exactly the extensivity wall behind the `sorry`s in `S1_60.lean`.